

Exchange-Reservoir and Engineered-Clock Audits for a Finite Completion Latch

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Abstract—We audit two proposed repairs of a finite execute-capture-renew history architecture. An excitation-exchange bath improves a previously tested full-transverse bath family, but none of five frozen finite-filter generators reaches retained-channel error 2^{-16} at the specified port deadline. Two perfect-transfer-inspired clock profiles also fail at the uniform clock's independently enclosed completion window, under identical resources, damping, and full-spectrum ceilings. These are conditional mathematical and simulation results; neither a microscopic bath certificate nor a physical two-local implementation is established.

Index Terms—Autonomous quantum control, Lindblad dynamics, reservoir engineering, finite-resource quantum clocks.

I. SCOPE AND FROZEN REQUIREMENTS

The Mayes Quantum Method (MQM) target is a finite quantum control architecture with complete output, donor, and spectator accounting. This successor preserves the parent thirteen-gate bank word and its failed full- X_R bath family. New profiles and exchange coupling were frozen before evaluation. Evaluation uses two distinct objects: an eight-dimensional admitted *port* model for bath testing, and a seventeen-state live *word* path for arrival testing. Evidence for the former is not transferred to the latter.

The bank contains $A, J, O_1, O_2, D_{1A}, D_{1J}, D_{2A}, D_{2J}, E$. Its word is

$$\begin{aligned} &H_A, T_A, \text{CNOT}_{AJ}, \text{SWAP}_{AO_1}, \\ &\text{SWAP}_{AD_{1A}}, \text{SWAP}_{JD_{1J}}, \\ &T_A, H_A, \text{CNOT}_{AJ}, \text{SWAP}_{AO_2}, \\ &\text{SWAP}_{AD_{2A}}, \text{SWAP}_{JD_{2J}}, X_E. \end{aligned}$$

A fresh flag F is distinct from exhaustion qubit E . The native clock has addresses $0, \dots, 13$, plus m, e, d ; an auxiliary qubit R mediates decay. These are abstract register specifications, not a physical qubit layout. Body background Hamiltonians vanish in this audit.

II. ENERGY BOOKKEEPING BEFORE BATH PROMOTION

Proposition 1 (resonant exchange). Let $[H_S, A_\nu] = -\nu A_\nu$ and $[H_B, b_\omega^\dagger] = \omega b_\omega^\dagger$. Then

$$[H_S + H_B, A_\nu b_\omega^\dagger] = (\omega - \nu) A_\nu b_\omega^\dagger. \quad (1)$$

Thus the interaction $A_\nu b_\omega^\dagger + \text{h.c.}$ conserves free energy exactly. By contrast, broadband $\sigma_R^- b_\omega^\dagger + \text{h.c.}$ conserves excitation count, but does not commute with free energy away from resonance. This distinction prevents an exchange approximation from

becoming an unsupported energy-conservation claim. Three integer-matrix checks verify resonant and detuned finite dilations.

The port interaction has hopping $h = g = 1$, auxiliary coupling $\lambda^2 = 4\sqrt{2}$, and reference damping $\kappa = 8$. Its laboratory Hamiltonian is $H_S = H_{\text{port}} + 64N$, where N counts the port/auxiliary excitation charge. The new bath hypothesis separates lowering and raising correlation blocks rather than coupling through full X_R . It retains the parent Ohmic spectrum:

$$\gamma_\downarrow(\omega) = \frac{\alpha\omega e^{-\omega/640}}{1 - e^{-\beta\omega}}, \quad \omega > 0, \quad (2)$$

$$\gamma_\uparrow(\omega) = e^{-\beta\omega} \gamma_\downarrow(\omega), \quad (3)$$

with $\beta = 5/16$ and $\alpha = 8(1 - e^{-20})e^{1/10}/64$. This defines the tested generator family; it does not provide a finite-time microscopic derivation.

For twelve positive Bohr frequencies ν_a , define separate positive Gram matrices

$$K_{ab}^s = \frac{\tau}{2\pi} \int_0^\infty \gamma_s(\omega) f_a(\omega) f_b(\omega) d\omega, \quad (4)$$

where $f_a(\omega) = \text{sinc}((\omega - \nu_a)\tau/2)$, $\text{sinc } x = \sin x/x$, and $s \in \{\downarrow, \uparrow\}$. Lowering operators use $K^\downarrow$ and their adjoints use $K^\uparrow$; cross-sign blocks vanish by the declared hypothesis. Positivity follows from the weighted squared-modulus integral. Finite filters still sample low-frequency thermal weight, so $e^{-\beta 64}$ alone does not bound the raising block.

III. RESERVOIR EVALUATION

Frozen $\tau \in \{1/64, 1/16, 1/4, 1, 4\}$ is tested at $t = 16$ with launch $pF0R0$ and target $dF1R0$. The bank is an identity spectator for this port-only test. Positive composite Gauss-Legendre quadrature of orders 16 and 32 agrees within 1.51×10^{-14} in matrix entries. This is numerical convergence, not a directed quadrature enclosure. Small negative floating eigenvalues below 2.4×10^{-14} in magnitude are clipped as recorded rounding errors. Every exchange case fails 1/65536. The best frozen case improves the parent by roughly 221 times, but remains approximately 1.86 times above target. A CPTP Duhamel comparison gives

$$\frac{1}{2} \|\Phi_t^K - \Phi_t^{K_0}\|_\diamond \leq \min\{1, t \sum_{ab} |\Delta K_{ab}| \|A_a\| \|A_b\|\}. \quad (5)$$

All evaluated bounds cap at one and are inconclusive. The inherited analytic omitted-tail allowance is 5102125/15199648742375424 for $\tau \leq 4$, $t \leq 17$, and cutoff

TABLE I
RETAINED PORT ERROR AT THE FROZEN DEADLINE

τ	Parent full- X_R	Exchange family
1/64	0.689243	0.00224227
1/16	0.307529	0.00307166
1/4	0.108757	0.000596812
1	0.0395829	0.000625790
4	0.00628457	0.0000283790

TABLE II
MATCHED COLD WORD ERROR AT $t = 1024$

Profile	$q(1024)$	Full interaction norm
Uniform	7.33968×10^{-6}	2.621372
PST16	4.30734×10^{-4}	2.431625
PST14	2.91352×10^{-4}	2.614782

48×640 . It bounds only that tail. Born, memory, Lamb-shift, and quadrature-enclosure debts remain open. Coarse-grained and secular approximations must be compared with a microscopic reference, not ranked universally [2].

IV. MATCHED CLOCK COMPARISON

Let $P_j = U_j \cdots U_1$, and extend P_{13} to m, e, d . The history gauge $V = \sum_j |j\rangle\langle j| \otimes P_j$ removes gate labels while preserving the whole bank and arbitrary references. Every dark branch carries the completed word. For fixed launch and the local cold bath, live and captured supports are orthogonal, giving exact retained half-diamond error

$$q(t) = \|e^{At}e_0\|^2, \quad A + A^T = -8|16\rangle\langle 16|. \quad (6)$$

Consequently $q'(t) \leq 0$. The real live generator has $A_{j,j+1} = -w_j$, $A_{j+1,j} = w_j$, and $A_{16,16} = -4$.

All profiles keep $w_{15} = \lambda$, $\kappa = 8$, and the same clock, flag, auxiliary, and bank. Uniform has $w_j = 1$ for $j < 15$. The two frozen candidates are

$$\text{PST16: } w_j = \sqrt{(j+1)(15-j)}/8, \quad j < 15, \quad (7)$$

$$\text{PST14: } w_j = \sqrt{(j+1)(13-j)}/7, \quad j < 13, \quad (8)$$

with PST14's marker and entry weights equal to one. The isolated transfer-chain construction is prior art [1]; here the dissipative terminal changes the problem. Denominators $8 = 2^3$ and 7 normalize each maximum edge to one. Their isolated transfer times 4π and $7\pi/2$ are not latch-completion deadlines.

Proposition 2 (common full-spectrum ceiling). The full 68-state scalar clock-flag-auxiliary interaction satisfies $\|H\| < 8/3$ for all three profiles. Its nonnegative bipartite forest is dominated entrywise by the uniform graph. The largest relevant component is the seventeen-site path; remaining components have norm at most two or λ . For $x = 8/3$, leading-minor recurrence $D_0 = 1$, $D_1 = x$, $D_{j+1} = xD_j - w_{j-1}^2 D_{j-1}$ is positive using $\lambda^2 < 198/35$, since $\sqrt{2} < 99/70$. Sylvester positivity and bipartite spectral symmetry prove the ceiling. With the same laboratory charge $64N$, the centered full-spectrum ceiling is $64 + 8/3 = 200/3$. These are matched ceilings, not equal measured norms.

An 80-digit directed interval calculation encloses e^{At} using degree-24 Taylor expansion, scale 2^{20} , analytic remainder, and

interval squaring. At $1024 - 1/1024$, uniform error is below $1/131072$. Both engineered errors exceed $1/65536$ even at $1024 + 1/1024$. Monotonicity certifies uniform success and both candidate failures throughout the same window. This rejects these profiles at this deadline, not all engineered clocks. The uniform threshold crossing near 918.897 is a simulation diagnostic, not an interval-certified minimum.

V. GOVERNANCE AND NEXT GATE

No hypothesis, threshold, or weight was changed after evaluation. A root finder initially assumed a crossing in $[0, 16384]$; failure handling now records “no bracketed crossing” without extending that frozen bracket. Additional engineered right-edge intervals certify whole-window rejection and are disclosed as follow-up diagnostics.

The resource ledger remains seventeen native clock levels, two control/auxiliary qubits, and nine bank qubits: full dimension $17 \times 2^{11} = 34816$. A five-bit address embedding is not a five-qubit two-local hardware implementation. Clock-controlled two-body bank gates generally remain three-subsystem terms; native joint admission, geometry, reservoir refresh, and physical validation remain open. Physical promotion is zero.

The highest-value next gate is a finite-time microscopic resonant-bath reference with an explicit error budget. In parallel, damping-aware clock engineering should optimize certified survival rather than isolated perfect transfer, with a new frozen development/evaluation split. No hardware advantage, threshold, or publication-level novelty is inferred from these audits.

REFERENCES

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